01 Errors in Measurements and Instruments

Error & Its Types

Whenever an experiment is performed, two kinds of errors can appear in the measured quantity:

(1) Random and

(2) Systematic errors

1. **Random errors** appear randomly because of operator, fluctuations in external conditions and variability of measuring instruments. The effect of random error can be some what reduced by taking the average of measured values. Random errors havse no fixed sign or size.
2. **Systematic errors** occur due to error in the procedure, or miscalibration of the instrument etc. Such errors have same size and sign for all the measurements. Such errors can be determined.

   A measurement with relatively small random error is said to have high precision. A measurement with small random error and small systematic error is said to have high accuracy.

   The experimental error [uncertainty] can be expressed in several standard ways.

Error limits $Q \pm \Delta Q$ is the measured quantity and ΔQ is the magnitude of its limit of error. This expresses the experimenter's judgment that the 'true' value of Q lies between $Q - \Delta Q$ and $Q + \Delta Q$. This entire interval within which the measurement lies is called the range of error. Random errors are expressed in this form.

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Absolute Error

Error may be expressed as absolute measures, giving the size of the error in a quantity in the same units as the quantity itself.

Least Count Error: If the instrument has known least count, the absolute error is taken to be equal to the least count unless otherwise stated.

Relative (or Fractional) Error

Error may be expressed as relative measures, giving the ratio of the quantity's error to the quantity itself.

In general,

$$\text{Relative error} = \frac{\text{Absolute error in a measurement}}{\text{Size of the measurement}}$$

We should know the error in the measurement because these errors propagate through the calculations to produce errors in results.

A. Systematic errors:

They have a known sign. The systematic error is removed before beginning calculations. Bench error and zero error are examples of systematic error.

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B. Random error:

They have unknown sign. Thus, they are represented in the form $A \pm a$.

Here we are only concerned with limits of error. We must assume a "worst-case" combination. In the case of subtraction, $A - B$, the worst-case deviation of the answer occurs when the errors are either $+a$ and $-b$ or $-a$ and $+b$. In either case, the maximum error will be $(a + b)$.

For example, in the experiment on finding the focal length of a convex lens, the object distance (u) is found by subtracting the positions of the object needle and the lens. If the optical bench has a least count of 1 *mm*, the error in each position will be 0.5 *mm*. So, the error in the value of u will be 1 *mm*.

Calculation of error in mathematical functions:

Addition and subtraction rule:

The absolute random errors add.

Thus, if $R = A + B$, $r = a + b$ and if $R = A - B$, $r = a + b$

Calculation of error in complex mathematical functions

Product and quotient rule: The relative random errors add.

Thus if $R = AB$, $\frac{r}{R} = \frac{a}{A} + \frac{b}{B}$ and if $R = \frac{A}{B}$, then also $\frac{r}{R} = \frac{a}{A} + \frac{b}{B}$

1. **Power rule:** When a quantity Q is raised to a power P, the relative error in the result is P times the relative error in Q. This also holds for negative powers. If $R = Q^P$, $\frac{r}{R} = P \times \frac{q}{Q}$

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2. The quotient rule is not applicable if the numerator and denominator are dependent on each other.

e.g if $R = \frac{XY}{X+Y}$. We cannot apply quotient rule to find the error in R. Instead we write the equation as follows $\frac{1}{R} = \frac{1}{X} + \frac{1}{Y}$. Differentiating both the sides, we get $-\frac{dR}{R^2} = -\frac{dX}{X^2} - \frac{dY}{Y^2}$.

Thus $\frac{r}{R^2} = \frac{x}{X^2} + \frac{y}{Y^2}$

Illustration 1

An experiment measures quantities x, y, z and then t is calculated from the data as $t = \frac{xy^2}{z^3}$. If percentage errors in x, y and z are respectively 1%, 3%, 2%, then find the percentage error in t.

Solution:

$$t = \frac{xy^2}{z^3}$$

$$\frac{\Delta t}{t} = \left|\frac{\Delta x}{x}\right| + 2\left|\frac{\Delta y}{y}\right| + 3\left|\frac{\Delta z}{z}\right| = 1 + 2(3) + 3(2)$$

$= 1 + 6 + 6 = 13\%$

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Illustration 2

A student finds the constant acceleration of a slowly moving object with a stopwatch. The equation used is $S = (1/2)AT^2$. The time is measured with a stopwatch, the distance, S with a meter stick. What is the acceleration and its estimated error?

$S = 2 \pm 0.005$ meter

$T = 4.2 \pm 0.2$ second

Solution:

We use capital letters for quantities, lower case for errors. Solve the equation for the result, a. $A = 2S/T^2$.

Its random error equation is $\frac{a}{A} = 2\frac{t}{T} + \frac{s}{S}$. Thus $A = 0.23 \pm 0.02\ m/s^2$.

Significant digits & rounding off

Significant digits

The reported result of measurement is a number that includes all digits in the number that are known reliably plus the first digit that is uncertain. The reliable digits plus the first uncertain digit are known as significant digits or significant figures.

Rules for significant figures in Computed value

The proper number of significant figures that a computed value should have is decided by a set of conventional rules. However, before we get to those rules for computed values we have to consider how to determine how many significant digits are indicated in the numbers being used in the math computation.

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A. Rules for determining the number of significant digits in number with indicated decimals.

1. All nonzero digits (1-9) are to be counted as significant.
2. Zeros that have any nonzero digits anywhere to the LEFT of them are considered significant zeros.
3. All other zeros not covered in rule (2) above are NOT be considered significant digits.

For example: 0.0040000

The 4 is obviously to be counted significant (Rule-1), but what about the zeros? The first three zeros would not be considered significant since they have no nonzero digits anywhere to their left (Rule-3). The last four zeros would all be considered significant since each of them has the nonzero digit 4 to their left (Rule-2). Therefore, the number has a total of five significant digits.

Here is another example: 120.00420

The digit 1, 2, 4 and 2 are all considered significant (Rule-1). All zeros are considered significant since they have non-zero digits somewhere to their left (Rule-2). So, there are a total of eight significant digits. If in the question, we are given a number like 100, we will treat that the number has only one significant digit by convention.

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B. Determining the number of significant digits if number is not having an indicated decimal.

The decimal indicated in a number tells us to what position of estimation the number has been indicated. But what about 1,000,000?

Notice that there is no decimal indicated in the number. In other words, there is an ambiguity concerning the estimated position. This ambiguity can only be clarified by placing the number in exponential notation.

For example:

If we write the number above in this manner.

1.00×10^6

We have indicated that the number has been recorded with three significant digits. On the other hand, if we write the same number as:

1.0000×10^6

We have identified the number to have 5 significant digits. Once the number has been expressed in exponential notation form then the digits that appear before the power of ten will all be considered significant. So, for example: 2.0040×10^4 will have five significant digits. This means that unit conversion will not change the number of significant digits.

Thus 0.000010 *km* = 1.0 *cm* = 0.010 *m* = 1.0×10^{-2} *m* = 1.0×10^{-5} *km*

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Rule for expressing proper number of significant digits in an answer from multiplication or division

For multiplication AND division there is the following rule for expressing a computed product or quotient with the proper number of significant digits.

The product or quotient will be reported as having as many significant digits as the number involved in the operation with the least number of significant digits.

For example: $0.000170 \times 100.40 = 0.017068$

The product could be expressed with no more that three significant digits since 0.000170 has only three significant digits, and 100.40 has five. So according to the rule the product answer could only be expressed with three significant digits. Thus, the answer should be 0.0171 (after rounding off)

Another example: $2.000 \times 10^4 / 6.0 \times 10^{-3} = 0.33 \times 10^7$

The answer could be expressed with no more that two significant digits since the least digit number involved in the operation has two significant digits.

Sometimes this would required expressing the answer in exponential notation.

For example: $3.0 \times 800.0 = 2.4 \times 10^3$

The number 3.0 has two significant digits and then number 800.0 has four. The rule states that the answer can have no more than two digits expressed. However, the answer as we can all see would be 2400. How do we express the answer 2400 while obeying the rules? The only way is to express the answer in exponential notation so 2400 could be expressed as: 2.4×10^3.

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Rule for expressing the correct number of significant digits in an addition or substraction:

The rule for expressing a sum or difference is considerably different than the one for multiplication of division. The sum or difference can be no more precise than the least precise number involved in the mathematical operation. Precision has to do with the number of positions to the RIGHT of the decimal. The more position to the right of the decimal, the more precise the number. So, a sum or difference can have no more indicated positions to the right of the decimal as the number involved in the operation with the LEAST indicated positions to the right of its decimal.

For example: 160.45 + 6.732 = 167.18 (after rounding off)

The answer could be expressed only to two positions to the right of the decimal, since 160.45 is the least precise.

Another example: 45.621 + 4.3 – 6.41 = 43.5 (after rounding off)

The answer could be expressed only to one position to the right of the decimal, since the number 4.3 is the least precise number (i.e. having only one position to the right of its decimal). Notice we aren't really determining the total number of significant digits in the answer with this rule.

Rules for rounding off digits:

There are a set of conventional rules for rounding off.

1. Determine according to the rule what the last reported digit should be.

2. Consider the digit to the right of the last reported digit.

3. If the digit to the right of the last reported digit is less than 5 round it and all digits to its right off.

4. If the digit to the right of the last reported digit is greater than 5 round it and all digits to its right off and increased the last reported digit by one.

5. If the digit to the right of the last reported digit is a 5 followed by either no other digits or all zeros, round it and all digits to its right off and if the last reported digit is odd round up to the next even digit. If the last reported digit is even then leave it as is.

For example, if we wish to round off the following number to 3 significant digits: 18.3682.

The last reported digits would be the 3. The digit to its right is a 6 which is greater than 5. According to the Rule-4 above, the digit 3 is increased by one and the answer is: 18.4.

Another example: Round off 4.565 to three significant digits.

The last reported digit would be the 6. The digit to the right is a 5 followed by nothing. Therefore, according to Rule-5 above since the 6 is even it remains so and the answer would be 4.56.

Illustration 3

Complete the following calculations, reporting your answer to the correct number of significant figures :

$$\frac{10,569-958.82-1.8\times10^{-1}}{0.9950+8.60+0.015}$$

Solution:

$$\frac{10,569-958.82-1.8\times10^{-1}}{0.9950+8.60+0.015}=1000$$

$$=1.00\times10^{3}$$

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10569	0.9950
−958.82	+8.60
−0.18	+0.015
9610.00	9.610
4 significant figure	3 significant figure

Illustration 4

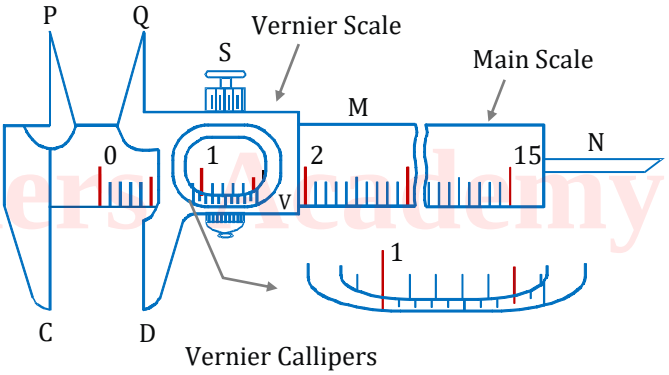
A student measuring the diameter of a pencil of circular cross-section with the help of a vernier scale records the following four readings 5.50 *mm*, 5.55 *mm*, 5.45 *mm*; 5.65 *mm*. The average of these four readings is 5.5375 *mm* and the standard deviation of the data is 0.07395 *mm*. The average diameter of the pencil should therefore be recorded as:

Ans. (5.54 ± 0.07) *mm*

Solution:

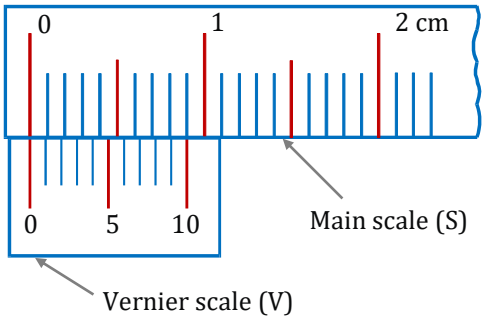
As all reading have only three *S.F.*, so value should also be reported upto three *S.F.*
Also, error will be upto two decimal places.

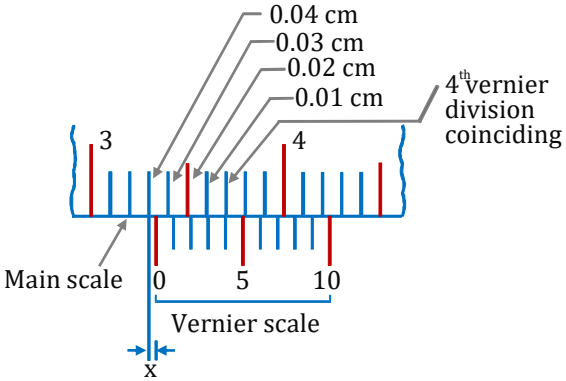
Vernier Callipers:



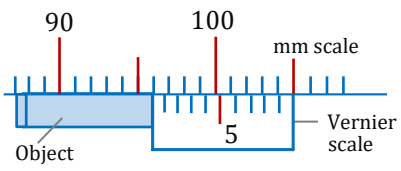
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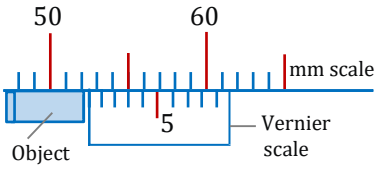
Principle of Vernier





Reading a vernier with 4th division coinciding





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It consists of a main scale graduated in cm/mm over which an auxiliary scale (or Vernier scale) can slide along the length. The division of the Vernier scale being shorter than the divisions of the main scale.

Least count of Vernier Callipers

The least count or Vernier constant (v. c) is the minimum value of correct estimation of length without eye estimation. If N division of vernier coincides with $(N\text{-}1)$ division of main scale, then

$$N\,(VS) = (N-1)\,ms \quad \Rightarrow \quad 1VS = \frac{N-1}{N}\,ms$$

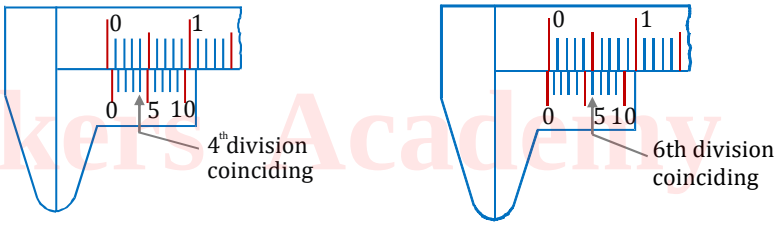
Vernier constant = $1\,ms - 1\,vs = \left(1-\frac{N-1}{N}\right)ms = \frac{1ms}{N}$, which is equal to the value of the smallest division on the main scale divided by total number of divisions on the vernier scale.

Length as measured by Vernier Callipers

The formula for measuring the length is L = main scale reading + least count of vernier scale × Vernier scale division coinciding with a main scale division

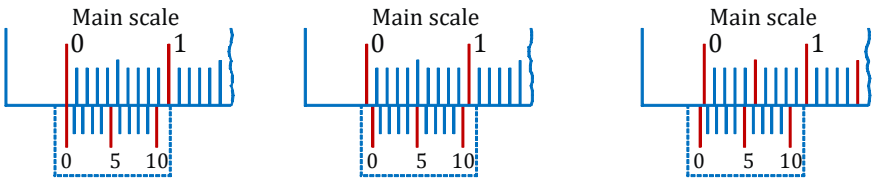
Main scale reading is given by the zeroth division of the vernier scale as shown in the figure.

Zero error:



Positive zero error (+0.04 cm) and its correction

Negative zero error= (–0.04 cm) and its correction



without zero error

with positive zero error

with negative zero error

(i)

(ii)

If the zero marking of main scale and vernier callipers do not coincide, necessary correction has to be made for this error which is known as zero error of the instrument.

Positive zero error:

If the zero of the vernier scale is to the right of the zero of the main scale the zero error is said to be positive and the correction will be negative.

The zero error is always subtracted from the reading to get the corrected value.

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Negative zero error:

If the zero of the vernier scale is to the left of the zero of the main scale the zero error is said to be negative and the correction will be positive.

If the zero error is positive, its value is calculated as we take any normal reading. If the zero-error negative (The zero of vernier scale lies to the left of the zero of main scale),

Negative zero error = – [Total number of *vsd* – *vsd* coinciding] × L.C.

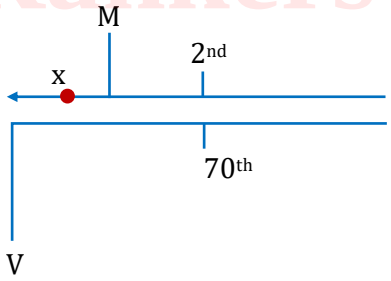
Do not try to read the main scale at the point where the lines match best. This has no meaning. Read from the vernier scale instead. Sometimes it is difficult to tell whether the best match of lines is for vernier marks 9, 0, or 1. Make your best estimate, but realize that the final result including the vernier must round off to the result you would choose if there was no vernier. If the mark is close to 3.20 on the main scale, but the vernier reading is 9, the length is 3.19 *cm*. If the mark is close to 3.2 on the main scale and the vernier is 1, the length is 3.21 *cm*.

Illustration 5

In a Vernier Callipers, the zero of vernier scale is left to the zero of main scale when both jaws are closely joined to each other. Total number of divisions on vernier scale is 80 which is equal to 2.32 cm. 70th division of vernier is exactly coinciding with 2nd division of main scale when jaws are closely joined to each other. Now diameter of cylinder is measured, it is found that zero of vernier lies between consecutive point 6.2 cm to 6.23 cm on the main scale and 50th division of vernier exactly coincides with a main scale. Then find the error of the venrier callipers.

Solution:

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Vernier Callipers has -ve zero error

1 main scale = 0.03

1 Vernier scale = 0.029

Let 'x' is the gap between zero of vernier scale and main scale

$$x+2\times0.03=70\times(0.029)$$

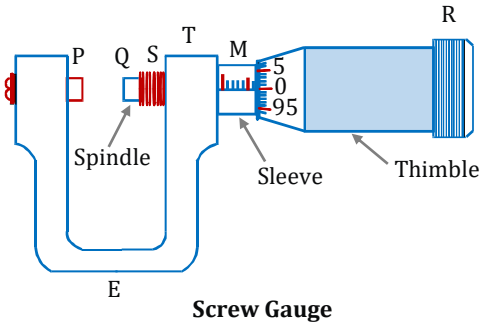
x = 1.97 cm

SCREW GAUGE

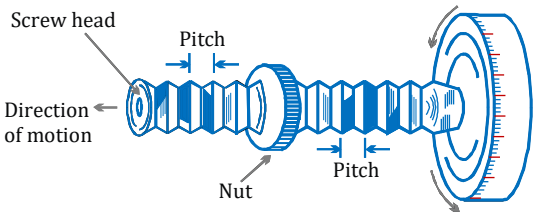
Screw Gauge (Micrometer Screw)

In general vernier callipers can measure accurately upto 0.01 *cm* and for greater accuracy micrometer screw devices e.g. screw gauge, spherometer are used. These consist of accurately cut screw which can be moved in a closely fitting fixed nut by turning it axially. The instrument is provided with two scales:

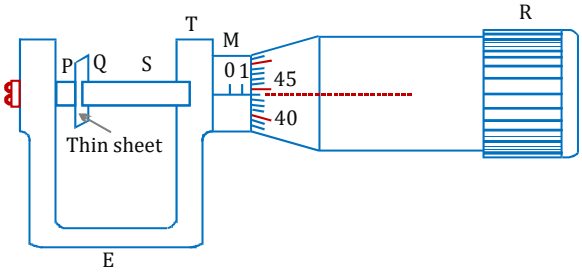
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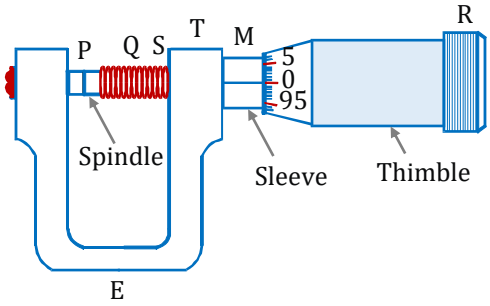


Screw Gauge

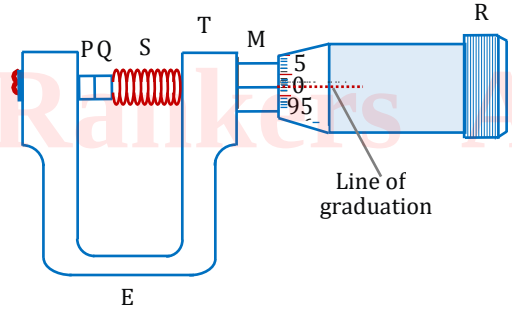


Principle of a micrometer

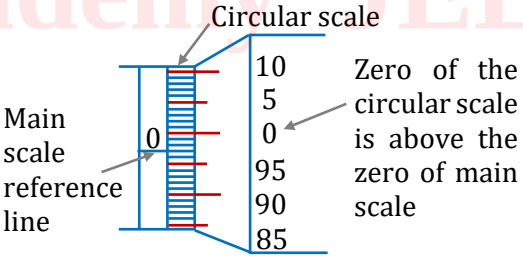


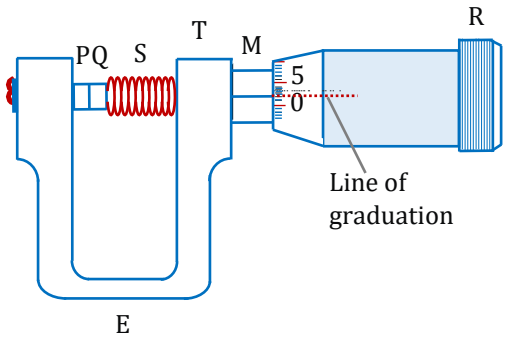


Screw gauge with no zero error

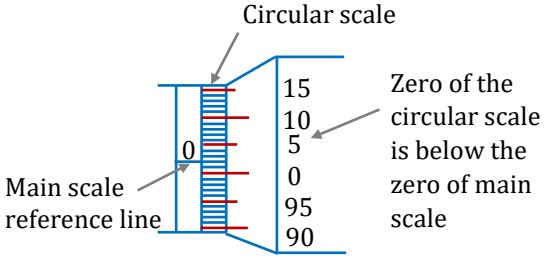


Negative zero error
(3 division error) i.e., - 0.003 cm





Negative zero error
(2 division error) i.e., + 0.002 cm



(i) The main scale or pitch scale *M* graduated along the axis of the screw.

(ii) The cap-scale or head scale *H* round the edge of the screw head.

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Constants of the Screw Gauge

(a) **Pitch:** The translational motion of the screw is directly proportional to the total rotation of the head. The pitch of the instrument is the distance between two consecutive threads of the screw which is equal to the distance moved by the screw due to one complete rotation of the cap. Thus for 10 rotation of cap = 5 *mm*, then pitch = 0.5 *mm.*

(b) **Least count:** In this case also, the minimum (or least) measurement (or count) of length is equal to one division on the head scale which is equal to pitch divided by the total cap divisions. Thus in the aforesaid illustration, if the total cap division is 100, then least count = 0.5*mm*/100 = 0.005 *mm.*

(c) **Measurement of length by screw gauge:**

$L = n \times$ pitch $+ f \times$ least count, where n = main scale reading and f = caps scale reading

Zero Error:

In a perfect instrument the zero of the head scales coincides with the line of graduation along the screw axis with no zero-error, otherwise the instrument is said to have zero-error which is equal to the cap reading with the gap closed. This error is positive when zero line or reference line of the cap lies **below** the line of graduation and versa. The corresponding corrections will be just opposite.

Illustration 6

When the gap is closed without placing any object in the screw gauge whose least count is 0.005 *mm*, the 5th division on its circular scale coincides with the reference line on main scale, and when a small sphere is placed reading on main scale advances by 4 divisions, whereas circular scale reading advances to five times to the corresponding reading when no object was placed. If there are 200 divisions on the circular scale, then find the radius of the sphere.

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Solution:

Error = 5 × *L.C.*

= 5 × 0.005 *mm*

MSR = 4 *mm*

VSR = 5 × (5 × 0.005) *mm*

Reading = *MSR* + *VSR* – Error

= 4*mm* + (20 × 0.005) *mm*

Reading = 4 *mm* + 0.1 *mm*

= 4.1 *mm*

This reading is measurement of diameter.

$$\text{Radius} = \frac{\text{Diameter}}{2} = \frac{4.1}{2}\text{mm}$$

Radius = 2.05 *mm*

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Searle's Experiment

EXPERIMENT

Objective: Determine Young's modulus of elasticity of the material of a given wire by using Searle's apparatus.

Apparatus required: - Searle's apparatus, a metre scale, a screw gauge, slotted weights, hanger etc.

Theory: Suppose a wire of length L and cross-section A is stretched by an amount ℓ by a force F acting along its length, then

$$\text{Stress} = \frac{F}{A} \text{ and strain} = \frac{\ell}{L}$$

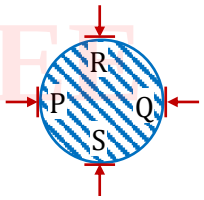
$$\therefore \quad \text{Young's modulus } Y = \frac{F}{A} \times \frac{L}{\ell}$$

Here $F = Mg$ and $A = \pi r^2$ $\Rightarrow \boxed{Y = \frac{MgL}{\pi r^2 \ell}}$

Procedure:

1. Attach weights to the hooks of both the frames so that the wires A and B become free from kinks and stretched. Then prepare it for the experimental load and unload one-kilogram weight on the hanger of the experimental wire B to remove its fatigue and attached only a constant weight W on the hook H_1 to keep the reference wire A taut.

2. Measure the length of the experimental wire from the point where it leaves the fixed support to the point where it is fixed in the frame. Find the pitch and least count of the screw gauge and determine the diameter of the wire at about ten different places and at each place in two mutually perpendicular directions. Determine the mean diameter and radius.



3. Consult the table of constants and find the breaking stress for the steel wire used. The maximum load should not exceed one-third of the breaking stress. Find the pitch and the least count of the micrometer attached to the frame and adjust it such that the bubble in the spirit level is exactly in the centre. Take the reading of the micrometer.

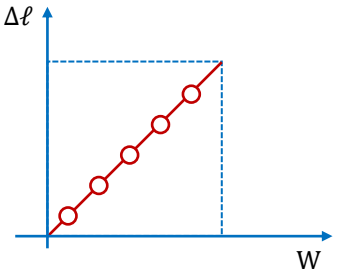
4. Now, increase gradually the load on the hanger of the experimental wire B in steps of 0.5 kg. Observe the reading on the micrometer at each stage, after levelling the instrument with the spirit level till the maximum permissible load is reached. In these cases, the adjusting screw has to be moved upwards to bring the bubble in the Centre. To avoid back lash error, all the final adjustments should be made by moving the screw in the upward direction only. If at any time the screw is raised too much, lower it below the central position and then raise it slowly to the proper position.

5. Unload the wire by removing the weights in the same order and take the reading on the micrometer screw each time. In this case the screw has to be moved downwards but again, to avoid the back-lash error, move the screw in the same direction only. These readings taken for a particular load while loading the wire or unloading the wire, should agree closely.

From the data obtained, a graph showing extension (Δl) against the load (W) is plotted which is obtained as a straight line passing through the origin.

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The slope of the line gives $\tan\theta = \frac{l}{W} = \frac{l}{Mg}$



Now, stress = $\frac{Mg}{\pi r^2}$ and strain = $\frac{l}{L}$

Y = Stress/ strain = $\frac{MgL}{\pi r^2 l} = \frac{L}{\pi r^2 \tan\theta}$

With known values of initial length L, radius r of the experimental wire and tanθ, Young's modulus Y can be calculated.

Precautions:

1. Both the wires should be supported from same rigid support.
2. Kinks should be removed from experimental wire before starting the experiment.
3. Diameter of wire should be measured at different places and along two mutually perpendicular directions at every place.
4. Slotted weights should be added and removed gently.
5. We should wait for taking observations after adding or removing weight.
6. Load should be increased or decreased in regular steps.
7. Do not load the wire beyond one-third of the breaking stress.
8. To avoid backlash error, the screw should always be turned in the same direction.

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Sources of Error:

1. Experimental wire may not have uniform cross-section through-out.
2. The slotted weights may not have standard weight.
3. The screw gauge may have friction or backlash error.

Illustration 7

In a Searle's experiment for determination of Young's Modulus, when a load of 50 kg is added to a 3 meter long wire micrometer screw having pitch 1 mm needs to be given a quarter turn in order to restore the horizontal position of spirit level. If its cross-sectional area is 10^{-5} m^2 then find the Young's modulus of the wire.

Solution:

Pitch = 1 mm

$\Delta\ell$ = Quarter turn = $\frac{\text{Pitch}}{4} = \frac{1}{4}$ mm

$\Rightarrow \Delta\ell = 0.25 \times 10^{-3}$ m

$\ell = 3m$, Load $(W) = 50\ kg$

$A = 10^{-5}\ m^2$

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$$Y=\frac{\frac{F}{A}}{\frac{\Delta\ell}{\ell}}=\frac{F\ell}{A\Delta\ell}=\frac{500\times3}{10^{-5}\times0.25\times10^{-3}}$$

$Y = 6 \times 10^{11}\ N/m^2$

Resonance column experiment

Golden Key Points:

1. Speed of sound in solids is given by, $v=\sqrt{\frac{Y}{\rho}}$ where Y - Young's modulus of elasticity p - density

2. Speed of sound in fluids is given by

$$v=\sqrt{\frac{B}{\rho}}$$

where B : Bulk modulus of elasticity of fluid

ρ : density of fluid

3. Newton's formula for speed of sound

$v=\sqrt{\frac{P}{\rho}}$ here, $B = P$ (assuming isothermal conditions)

4. Laplace formula for speed of sound

$v=\sqrt{\frac{\gamma P}{\rho}}$ here $B = \gamma P$ assuming adiabatic conditions.

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5. At constant temperature, speed of sound is independent of pressure and density.

6. Effect of temperature on speed of sound

$v \propto \sqrt{T}$ or $\boxed{v_t = v_0 + 0.61t}$

Resonance Tube:

It has a 100 *cm* long glass or brass tube *AB* of about 2.5 *cm* internal diameter. The tube is fixed on a vertical board along the side of a metre scale marked in millimeters. The zero of the scale coincides with the upper end of the tube. The lower end of the glass tube is drawn out and is connected to a reservoir of water with the help of rubber tubing. A pinch cock is attached with the rubber tubing. The water level in the resonance tube *AB*, can be adjusted by manipulating the adjustable screw attached with the reservoir for clamping. To find adjustments of the water level in the tube *AB*, the pinch-cock is used. The tube *AB* can be made vertical with the help of the levelling screws *SS* provided at the heavy base of the frame.

End correction:

It is found that antinode occurs slightly above the open end of the resonance tube. So, a small correction has to be applied to the observed resonant length of the air column. This correction is called end correction. Theoretically, it was found by lord Rayleigh and it is 0.3*D* or 0.6 *R*, where *D* is diameter and *R* is radius of the tube.

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Experiment:

Objective: To find the speed of sound in air at room temperature using a resonance tube by two resonance positions.

Apparatus required: Resonance tube apparatus, tuning forks of different frequencies, a rubber pad, a thermometer, a set square, a spirit level etc.

Concept / Theory:

If ℓ_1 and ℓ_2 are the lengths of the air columns for the first and the second positions of resonance respectively,

$\ell_1 + x = \ell/4$ ...(i)

and $\ell_2 + x = 3\ell/4$ ...(ii)

x = end correction

ℓ = wavelength of the sound wave

So, $\ell_2 - \ell_1 = \frac{\lambda}{2}$

or $\ell = 2(\ell_2 - \ell_1) \quad \Rightarrow \because v = n\ell$

where v and n are the velocity and frequency of the sound wave respectively.

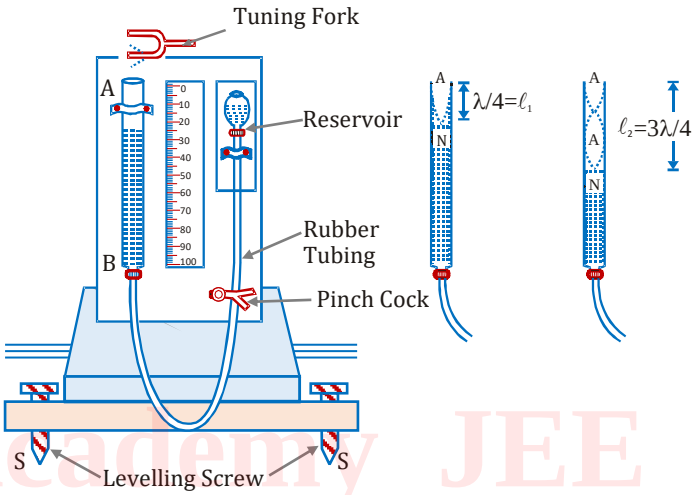
So, $v = 2n(\ell_2 - \ell_1)$

The velocity at $0°C$ is given by,

$v_0 = (v_t - 0.61 \times t)$ in ms^{-1}

where v_t is velocity at room temperature $t°C$.

Also, by eliminating ℓ from equations (i) and (ii), the end correction.



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Procedure:

1. With the help of the plumb line and the levelling screws at the base set the resonance tube AB vertical. Fill water in the reservoir R and some portion of the resonance tube.
   Suspend the thermometer by the side of the resonance tube to note the room temperature.
2. Release the pinch-cock and adjust the level of water in the resonance tube near the end A by adjusting the position of the reservoir R and then, after closing the pinch-cock lower down the position of the reservoir R.
3. Strike the tuning fork gently on the rubber pad and place it just above the upper end A of the resonance tube so that the prongs of the vibrating tuning fork are in vertical plane. Now open then pinch-cock and let the water level fall in the resonance tube slowly. At some position of the water level, you will listen sound of increasing loudness.
4. Repeat above steps to get the exact position of water level in the resonance tube for which the sound is of maximum intensity. Note the position of the water level with the help of a set square keeping one of its perpendicular edges tangential to meniscus of water and other edge parallel to line of graduations on the metre scale. Note the length ℓ_1 of the resonance column as shown in figure. This position corresponds to the first resonance position. Confirm the resonance positions

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by taking four readings, two when the level of water is falling and the other two when the water level is rising. Note down these lengths of the air column as ℓ_1.

5. Lower the position of water level so that it is increased about three times the length ℓ_1. Repeat above steps to get the second position of resonance with the same tuning fork. Note this length ℓ_2 of air column.

Errors:

The major systematic errors introduced are due to end effects in (end correction) and also due to excessive humidity.

Random errors are given by $\frac{\delta C}{C}=\frac{\delta(l_2-l_1)}{l_2-l_1}=\frac{\delta l_2+\delta l_1}{l_2-l_1}$.

Precautions:

1. The resonance tube should be set vertical using leveling screw.
2. Tuning fork should be vibrated gently by a rubber pad.
3. Prongs should be vibrated in a vertical plane just above the mouth (end) of the resonance tube.
4. Readings should be taken for water level rising as well as water level falling in the tube.
5. While measuring air temperature, thermometer bulb should not touch water or sides of resonance tube.

Sources of error:

1. Frequency of the tuning fork may differ than prescribed.
2. Resonance tube may not be vertical.
3. Thermometer may have faulty calibration.
4. There may be humidity in air column which can affect sound's velocity.
5. Meter scale attached with the resonance tube may have faulty calibrations.

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OHM'S LAW EXPERIMENT

Ohm's law:

According to this law the ratio of the potential difference across any conductor to the electric current flowing through it is constant, under constant physical conditions i.e. temperature, pressure etc. If V is the potential difference between the two points along a conductor and I is current flowing through it, then $V = IR$. Where R is constant. This constant is known as resistance of the conductor.

Resistance:

Obstruction in the flow of charge through a conductor is called a resistance. Its S.I. unit is Ohm (Ω).

Conductance:

The inverse of resistance is known as conductance. Its S.I. unit is $(ohm)^{-1}$ or mho or siemen.

Specific resistance or resistivity:

For conductors of given material at given temperature, the resistance is directly proportional to the length ℓ and inversely proportional to the area of cross-section A, i.e.,

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$$R \propto \frac{\ell}{A} \Rightarrow R = \frac{\rho \ell}{A}$$

where ρ is a constant which depends upon the nature of the material and is called resistivity or specific resistance of the conductor. For unit length and unit area of cross-section, ρ = *R*; so specific resistance is the resistance of a conductor of unit length and of unit area of cross-section. Its S.I. unit is ohm-metre.

Temperature dependence of resistivity:

When temperature of a conductor increases, ρ increases as; $\rho_\theta = \rho_0 (1 + \alpha \theta)$. Resistance of metals increase with rise in temperature. $R_\theta = R_0 (1 + \alpha \theta)$, where α is coefficient of resistance given as

$$\alpha = \frac{R_\theta - R_0}{R_0 . \theta} (^\circ C)^{-1}$$

Electrical conductivity:

Electrical conductivity is the inverse of term resistivity and is denoted by sigma (σ) given, as,

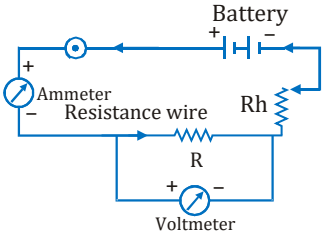
$$\sigma = \frac{1}{\rho} = \frac{\ell}{RA}$$ (S.I. unit of σ is $ohm^{-1}m^{-1}$)

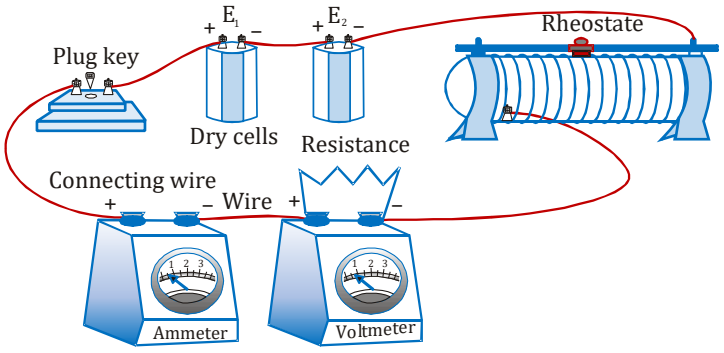
Experiment:

Objective: To determine the resistance of a given wire by plotting a graph of potential difference versus current.

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Apparatus required: A given resistance wire in the form of a coil, a battery eliminator or an accumulator or two dry cells, a *dc* voltmeter, a *dc* ammeter, a rheostat, one plug key, thick connecting wire, sand paper etc.





Circuit Diagram:

Procedure:

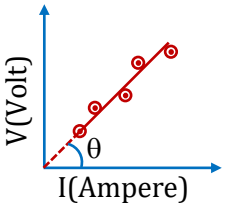
1. Draw the circuit diagram as shown in figure.
2. Note the range, the least count, and the zero error of voltmeter as well as that of the ammeter.
3. Insert the plug-in key *K*. Slide the rheostat contact to the extreme right figure. So that the current passing through the resistance wire is minimum.

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4. Adjust the rheostat and record the readings of the ammeter and the voltmeter. Then shift the rheostat contact to increase the current and note readings again. Take similarly ten observations.
5. Cut the resistance wire at the ends just coming out of voltmeter and measure its length 'ℓ', using meter scale.

Graph:

Using the readings of voltmeter (V) and ammeter (I) draw a graph as straight line best fitting all the points.



Calculation:

Slope of $V-I$ curve = $\frac{\Delta V}{\Delta I} = \tan\theta = R$

Error analysis:

The error in computing the ratio $R = \frac{V}{i}$ is given by $\left|\frac{\delta R}{R}\right| = \left|\frac{\delta V}{V}\right| + \left|\frac{\delta i}{i}\right|$

where δV and δi are of the order of the least counts of the instruments used.

Precautions:

1. Connections should be neat, clean and tight.
2. Connecting wires should be of thick copper wires.
3. Voltmeter and ammeter should be of proper range.
4. The key/plug should be inserted only while taking observations, otherwise current gives unnecessary heating of wires.
5. The unknown resistance should not be too low (than internal resistance of battery).

Sources of error:

1. Connections may be loose.
2. Rheostat may have very high resistance.
3. The unknown resistance may be too low.
4. Thick connecting wires may not be available.

Meter bridge experiment:

Wheatstone Bridge:

Wheatstone bridge is a special electric network in which four resistors P, Q, R and X are connected to form the four arms of a quadrilateral $ABCD$. A cell with key K_1 is connected across the diagonal AC and a galvanometer with key K_2 across another diagonal BD. The arrangement may be used to measure resistance of a conductor accurately.

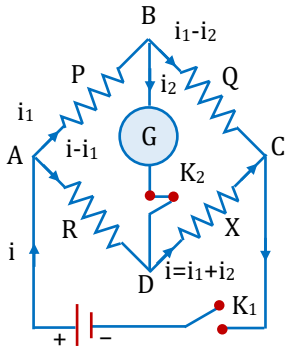
The resistances P and Q are kept in a fixed ratio and hence they are said to form the ratio arms of the bridge. Unknown resistance is placed for resistance X and R is a resistance box to provide variable known resistance in steps.

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After closing keys K_1 and K_2 resistance in resistance box R may be adjusted to obtain null deflection in galvanometer. In this situation bridge is said to be balanced. When bridge is balanced, B and D must be at same potential.

$$V_B = V_D$$



No current flows through the galvanometer i.e., $i_2 = 0$

$\therefore$ $V_A - V_B = V_A - V_D$ and $V_B - V_C = V_D - V_C$

$\Rightarrow$ $Pi_1 = R(i - i_1)$ ...(i)

and $Q(i_1 - i_2) = X(i - i_1 + i_2)$ ...(ii)

$Qi_1 = X(i - i_1)$ ...(iii)

Dividing (i) and (iii), we get $\frac{P}{Q} = \frac{R}{X}$

This, is the condition of balanced bridge. [Wheatstone bridge]

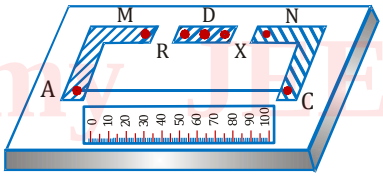
In balancing condition of the bridge, unknown resistance can be calculated.

Two practical versions of Wheatstone bridge are:

(a) Post office box, and

(b) Meter bridge or slide wire bridge.

Meter Bridge:

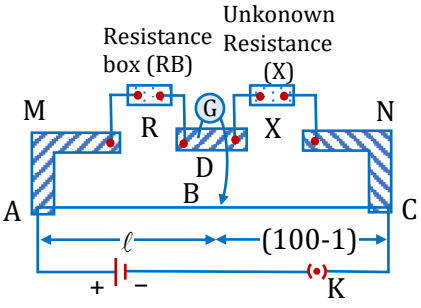


A meter bridge consists of a one-meter long wire AC, made of a homogeneous material constantan or eureka alloy and having a uniform cross-section. The wire is stretched on a wooden board. A meter scale is fixed on the board parallel to the wire.

The ends A and C of the wire are fixed to two thick copper strips M and N respectively. Another strip D is fixed between strips M and N to form two gaps for introducing resistances R and X. Connecting terminals are attached at the ends of the copper strips and also in the middle of strip D for making connections.

Experiment:

Objective: To find resistance of a given wire using a meter bridge and hence determine the specific resistance of its material.



Apparatus required: A resistance box, a wire about 1 meter long (of the material whose specific resistance is to be determined), a meter bridge, a jockey, one- way key, a galvanometer, a battery eliminator or a cell, thick connecting wires, sand paper, screw gauge etc.

Required formula: Unknown resistance $X = R\left(\frac{100-\ell}{\ell}\right)$ and specific resistance of the material of the given wire, $\rho = \frac{XA}{L} = \frac{X(\pi r^2)}{L}$, where r and L are the radius and length of the given wire respectively.

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Procedure:

1. Arrange the meter bridge and the various component as shown in figure and make tight connections.

   Putting *RB* in right gap and *X* in left gap.
2. Take out the plug from *RB* to introduce a suitable resistance say, $R = 2\Omega$ and close the key *K*.
3. Now, touch the jockey on meter-bridge wire at different places to obtain such a position, where there is no deflection in the galvanometer. At this condition

   $AB = \ell$ and $BC = (100 - \ell)$

   Putting *RB* in left gap and *X* in right gap.
4. Repeat step 2 and 3. But in this balancing condition, $BC = \ell$ and $AB = (100 - \ell)$.
5. Now disconnect the unknown resistance wire from the circuit. Straighten it by stretching and remove any kinks and measure its length with the help of a meter scale.
6. Measure the diameter of the wire with the help of a screw gauge.

Errors:

The major systematic errors in this experiment are due to the heating effect, end corrections introduced due to shift of the zero of the scale at *A* and *B*, and stray resistances in *P* and *Q*, and errors due to non-uniformity of the meter bridge wire.

Error analysis:

End corrections can be estimated by including known resistances P_1 and Q_1 in the two ends and finding the null point:

$$\frac{P_1}{Q_1} = \frac{l_1 + \alpha}{100 - l_1 + \beta} \quad \ldots(1)$$

where α and β are the end corrections.

When the resistance Q_1 is placed in the left gap and P_1 in the right gap,

$$\frac{Q_1}{P_1} = \frac{l_2 + \alpha}{100 - l_2 + \beta} \quad \ldots(2)$$

which give two linear equation for finding α and β.

In order that α and β be measured accurately, P_1 and Q_1 should be as different from each other as possible.

For the actual balance point, $\frac{P}{Q} = \frac{l + \alpha}{100 - l + \beta} = \frac{l_1'}{l_2'}$

Errors due to non-uniformity of the meter bridge wire can be minimized by interchanging the resistances in the gaps *P* and *Q*.

$$\therefore \quad \frac{\delta P}{P} = \left|\frac{\delta l_1'}{l_1'}\right| + \left|\frac{\delta l_2'}{l_2'}\right|$$

where, $\delta l'_1$ and $\delta l'_2$ are of the order of the least count of the scale.

The error is, therefore, minimum if $l'_1 = l'_2$ i.e. when the balance point is in the middle of the bridge. The error in ρ is $\frac{\delta\rho}{\rho} = \frac{2\delta r}{r} + \frac{\delta L}{L} + \frac{\delta P}{P}$.

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Precautions:

1. The connections should be neat, clean and tight.
2. All the plugs in the resistance box should be tight.
3. While moving the jockey to locate the balance point, the jockey should be lifted again and again and should not be pressed and slided on the wire throughout.
4. The plug-in key K should be inserted only when the observations are to be taken.
5. The balance point should always be obtained near the midpoint or in between from 30 *cm* to 70 *cm*.
6. Diameter of wire should be measured in two mutually perpendicular direction at one place.

Source and of error:

1. The plugs may not be clean.
2. The meter bridge wire may not have uniform cross-section.
3. Connections may loose.
4. The screw gauge may have faulty calibration or backlash error.

Potentiometer experiment:

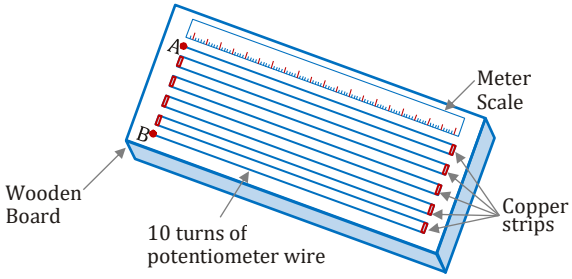
Potentiometer:

1. It is a device used to measure potential difference between two point in an electric circuit or emf of a cell with greater accuracy.
2. It behaves like an ideal voltmeter means its effective resistance is infinity.
3. It uses the concept of zero deflection during measurement.

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Construction:

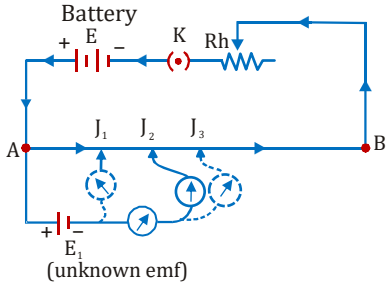
A thin, long, straight and uniform wire AB of constantan or manganin, which has a low temperature coefficient of resistance and high resistivity, is stretched on a wooden board along a meter-scale. The longer wire (taken long to increase accuracy) is arranged in the form of parallel wires and its ends are connected by thick strips of copper, so that the current flows through the wires in series.



Principle:

When a steady current flow through the potentiometer wire from the end A to B, as shown in figure, by a battery E then an electric potential fall uniformly along the whole length of the wire. This fall in potential per unit length of the wire, is known s "potential gradient".

Due to the constant current through potentiometer wire potential difference between any two points on the wire is directly proportional to the distance between the points, means



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$V \propto \ell$

$\Rightarrow \quad V = x\ell \quad$ or $\quad \frac{V}{\ell} = x$ = potential gradient

By comparing the unknown emf, say as E_1 in the figure, with the help of a galvanometer to the uniformly distributed potential over the wire, we can find the value of unknown. So, now suppose the jockey is made to touch a point J_1 on the wire such that the potential difference between A and J_1 is lower than the emf of E_1, then a resultant current flow through the secondary circuit or galvanometer and the needle of the galvanometer deflects in one direction.

On the other hand, if the jockey is made to touch a point J_2 on the wire such that the potential difference between A and J_2 is higher than the emf of the cell E_1, then a resultant current flow in opposite direction than previous one in the galvanometer and its needle deflects in the opposite side.

It is evident that in between J_1 and J_2, there will be a point J such that when the jockey is made to touch on J, there will be no deflection in the galvanometer. This point J is called **balancing null point** and the distance AJ is called **balancing length.** At this situation.

p.d. between A and J = emf of the cell

$\Rightarrow \quad x\ell = E_1 \quad \Rightarrow \quad E_1 = x\ell$

where x - potential gradient ; ℓ - balancing length ; E_1 - unknown emf

Sensitivity of Potentiometer:

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If on displacing the jockey slightly from the null point position, the galvanometer shows a large deflection, then the potentiometer is said to be sensitive. The sensitivity of the potentiometer depends upon the potential **gradient as:**

$$\text{Sensitivity of potentiometer} \propto \frac{1}{\text{Potential gradient}}$$

The potential gradient can be reduced by employing a long wire in the potentiometer.

Experiment

Objective: To compare the emf of two given primary cells (Daniel and Lechlanche Cell) using a potentiometer.

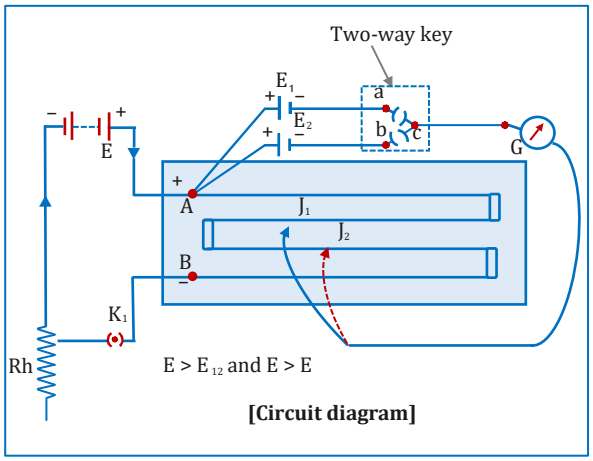
Apparatus required: A battery eliminator, a low resistance box about 20Ω, a one-way key, two primary cells (Daniel and Lechlanche), one two-way key, a resistance box plug type, a galvanometer, a potentiometer with a sliding jockey, connecting wires etc.

Formula required:

where E_1 and E_2 are the emf's of the given primary cells and ℓ_1 and ℓ_2 are their respective balancing lengths on potentiometer wire.

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Circuit diagram:



[Circuit diagram]

Procedure:

1. Arrange the circuit as shown in figure and plug key K_1.
2. In this circuit, by inserting the plug in the gap ac, cell E_1 comes in the secondary circuit. Introduce minimum resistance by *Rh* (rheostate) and then press the jockey at one end of the wire and note the direction of deflection in the galvanometer. Now repeat the same process with the jockey pressed near the other end of the wire and note the direction of deflection. The connections are correct if two deflections are in opposite directions.
3. If deflections are obtained in same direction, check connections and assure that $E > E_1$ and $E > E_2$.
4. After this, touch the jockey at different places on the wire and obtain a balance point, where on pressing the jockey, there is no deflection in the galvanometer note the length AJ_1 as ℓ_1.
5. Introduce E_2 in the *S* circuit by transferring the plug *ac* to *bc*. Obtain the balancing point J_2 and note the length AJ_2 as ℓ_2.
6. Repeat the experiment by shifting the contact point of rheostate.

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Precautions:

1. The plugs should be introduced in the keys only when the observations are to be taken.
2. The positive poles of the battery and cells E_1 and E_2 should all be connected to the terminal at zero of the wires.
3. The emf of the battery should be greater than the emfs of either of the two cells.
4. The jockey should not be rubbed along the wire. It should touch the wire gently.
5. The current should not be passed for a long time so as to avoid any heating of the wires resulting into change of resistance.
6. The balancing point should always be obtained at large distance and if possible, it should be obtained at the last wire.

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Source of error:

1. The battery (E) may not be fully charged.
2. By rubbing the jockey on the wire, it may not have uniform cross-section throughout its length.
3. The resistance of copper strips may not be zero.
4. Due to the current passed through the wire for a long time, its resistance may change.

Post office box experiment

Measurement of unknown resistance using a P.O. Box

A P.O. Box can also be used to measure an unknown resistance. It is a Wheatstone Bridge with three arms P, Q and R; while the fourth arm(s) is the unknown resistance. P and Q are known as the ratio arms while R is known at the rheostat arm. At balance, the unknown resistance

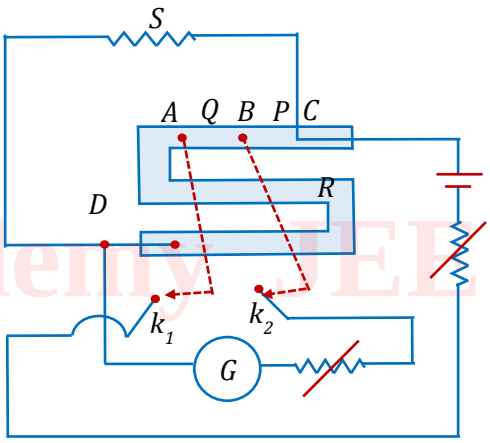
$$S = \left(\frac{P}{Q}\right) R \qquad \ldots(1)$$

The ratio arms are first adjusted so that they carry 100 Ω each. The resistance in the rheostat arm is now adjusted so that the galvanometer deflection is in one direction, if $R = R_0$ (Ohm) and in the opposite direction when $R = R_0 + 1$ (ohm).

This implies that the unknown resistance, S lies between R_0 and $R_0 + 1$ (ohm). Now, the resistance in P and Q are made 100 Ω and 1000 Ω respectively, and the process is repeated.

Equation (1) is used to compute S.

The ratio P/Q is progressively made 1 : 10, and then 1 : 100. The resistance S can be accurately measured.



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Errors:

The major sources of error are the connecting wires, unclear resistance plugs, change in resistance due to Joule heating, and the insensitivity of the Wheatstone bridge.

These may be removed by using thick connecting wires, clean plugs, keeping the circuit on for very brief periods (to avoid Joule heating) and calculating the sensitivity.

In order that the sensitivity is maximum, the resistance in the arm P is close to the value of the resistance S.

Half deflection method

Galvanometer:

- It is a device which is used to detect the presence of current or to measure weak current in the circuit.
- It is based on the principle that a current carrying coil placed in a uniform magnetic field experiences a torque.
- Basically, galvanometers are of two types:
  (i) Moving coil type (ii) Moving Magnet type (Tangent Galvanometer)

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Principle of a moving coil galvanometer

When a current I is flowing through a coil in a uniform magnetic field, the coil experiences first deflecting torque $\tau = NIAB$ and then restoring torque $\tau = C\theta$. Because both the torque opposes each other, therefore in equilibrium.

$$\tau_{\text{deflecting}} = \tau_{\text{restoring}}$$

$$NIAB = c\theta$$

$$\Rightarrow \quad I = \left(\frac{C}{BAN}\right)\theta = k\theta$$

$$\Rightarrow \quad I \propto \theta \quad \text{or} \quad \text{Current} \propto \text{deflection} \quad \text{where } k = \frac{C}{BAN} \quad \text{(Galvanometer's constant)}$$

i.e. deflection of the needle in the galvanometer is directly proportional to the current flowing through it.

(i) Figure of Merit: Current required to produce a deflection of one division is called figure of merit of galvanometer.

$$F = \frac{I}{\theta} = k$$

(ii) Current Sensitivity: Deflection per unit current is called current sensitivity of galvanometer.

$$S_C = \frac{\theta}{I} = \frac{1}{k} = \frac{1}{F} = \text{reciprocal of figure of merit}$$

(iii) Voltage sensitivity: Deflection per unit voltage i.e.

$$S_V = \frac{\theta}{V} = \frac{\theta}{IR} = \frac{S_C}{R}$$

Special Note:

Suppression type moving coil galvanometers are used in advanced laboratories in colleges and universities for very accurate measurements whereas in school laboratories pivoted type (weston) galvanometers are used.

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Experiment

Objective:

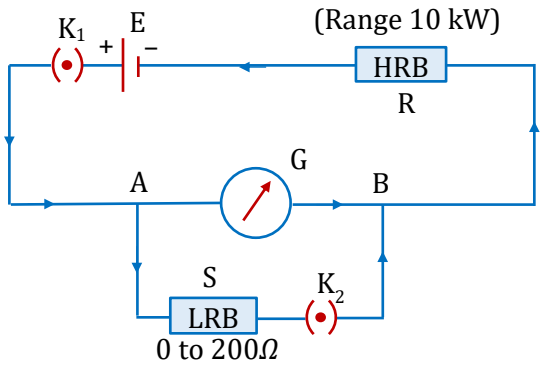
(a) To determine the resistance of a galvanometer by half deflection method and

(b) To find its figure of merit.

Apparatus required:

A galvanometer, a battery or accumulator, a low resistance box (*LRB*), a high resistance box (*HRB*), two one-way keys, connecting wires etc.

Circuit diagram:



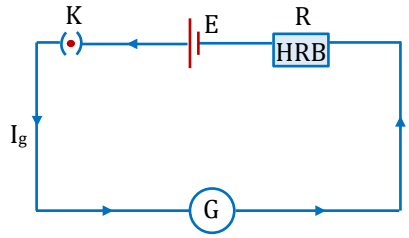
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Theory:

(a) Resistance of galvanometer by half deflection method: When a high resistance R is applied in the circuit with K_1 closed and K_2 open, the galvanometer draws a current I_g and shows a deflection θ such that

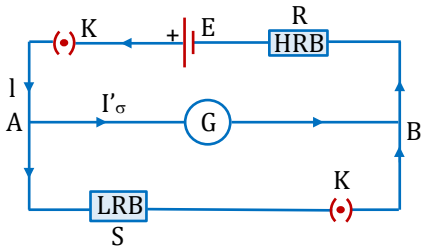
$$I_g = \frac{E}{R+G} \quad \text{...(i)}$$

where E, is emf of the battery and G is resistance of the galvanometer.



Now K_2 is closed. Adjust the resistance in LRB such that galvanometer deflection becomes equal to. $\frac{\theta}{2}$ Now the galvanometer draw the current.

$$I'_g = \frac{IS}{G+S} = \frac{ES}{R(G+S)+GS} \quad \left(\text{where } I = \frac{E}{R+\frac{GS}{G+S}}\right)$$



Also $I'_g = \frac{1}{2} I_g$

$$\therefore \quad \frac{ES}{R(G+S)+GS} = \frac{1}{2}\cdot\frac{E}{R+G}$$

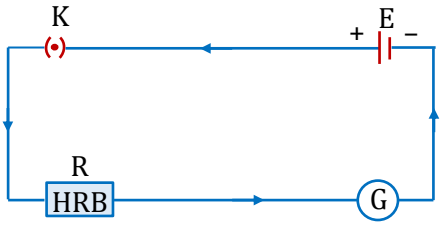
Gives $G = \frac{RS}{R-S}$

Knowing R and S, G can be calculated.

Also, if $R >> S$, S can be dropped in comparison to R and then $\boxed{G \simeq S}$.

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(b) Figure of merit of galvanometer is defined as the current required per division of deflection in galvanometer. It is denoted by k. Figure merit $k = \frac{I}{\theta}$

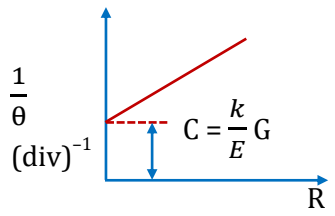


The circuit diagram for determination of figure of merit (k) of a galvanometer is shown in the figure.

When a high resistance R is introduced in the circuit through HRB, a small current I_g is drawn by it and it shows a deflection θ such that

$$I_g = k\theta = \frac{E}{R+G}$$

$\Rightarrow$ Figure of merit $k = \frac{1}{\theta}\left(\frac{E}{R+G}\right)$ ...(ii)



(Graph between (1/θ) and R)

Maximum current measured by galvanometer or full-scale deflection current for galvanometer

I_g = Number of division on one side of galvanometer scale × figure of merit

From equation (ii)

$$\frac{1}{\theta} = \frac{k}{E}R + \frac{k}{E}G \quad \text{...(iii)}$$

$\therefore$ Graph between $\frac{1}{\theta}$ and R is as shown.

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Procedure:

1. In the circuit, introduce a high resistance R in HRB and then insert the plug in the key K_1. Adjust the value of R to get deflection θ in even number of divisions.
2. Now close the key K_2 also and adjust the shunt resistance S from low resistance box (LRB) to get a deflection exactly half of θ in galvanometer.
3. Repeat experiment for different values of R and θ.

Precautions:

1. All connections in the circuit should be neat, clean and tight.
2. All the plugs in HRB and LRB should be tight.
3. A very high resistance R from HRB should be introduced first and then key K_1 should be closed to avoid any over current damage in galvanometer.
4. The emf of the cell or battery should be constant.
5. The deflection in the galvanometer should be as large as possible and should be even number of divisions.

Sources of error

1. The plugs of HRB and LRB may not be clean.
2. The emf of the battery may not be constant.
3. The galvanometer divisions may not be uniform.

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U-V Method:

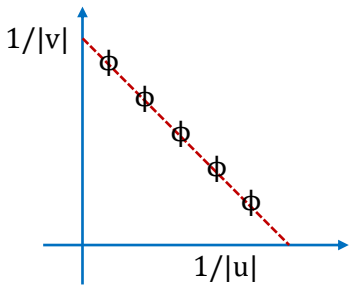
Focal length of a concave mirror and a convex lens using the *u-v* method.

In this method one uses an optical bench and the convex lens (or the concave mirror) is placed on the holder.

The position of the lens is noted by reading the scale at the bottom of the holder. A bright object (a filament lamp or some similar object) is placed at a fixed distance (u) in front of the lens (mirror).

The position of the image (v) is determined by moving a white screen behind the lens until a sharp image is obtained (for real images).

For the concave mirror, the position of the image is determined by placing a sharp object (a pin) on the optical bench such that the parallax between the object pin and the image is nil.



A plot of $|u|$ versus $|v|$ gives a rectangular hyperbola. A plot of $\frac{1}{|v|}$ vs $\frac{1}{|u|}$ gives a straight line. The intercepts are equal to $\frac{1}{|f|}$, where f is the focal length.

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Error:

The systematic error in this experiment is mostly due to improper position of the object on the holder. This error maybe eliminated by reversing the holder (rotating the holder by 180° about the vertical) and then taking the readings again. Averages are then taken.

The equation for random errors gives:

$$\frac{\delta f}{f^2} = \frac{\delta u}{u^2} + \frac{\delta v}{v^2}$$

The errors δu, δv correspond to the error in the measurement of u and v. Actually, we know the errors in the object position, lens position & image position. So, the errors in u & v are too be estimated as described before.

Index Error or Bench Error and its correction:

In an experiment using an optical bench we are required to measure the object and image distances from the pole or vertex on the mirror. The distance between the tip of the needles and the pole of the mirror is the actual distance. But we practically measure distances between the indices with the help of the scale engraved on the bench. These distances are called the observed distances. The actual distances may not be equal to the observed distances and due to this reason an error creeps in the measurement of the distances. This error is called the index or the bench error. This error is estimated with the help of a needle of known length placed horizontally between the tip of the needle and the pole.

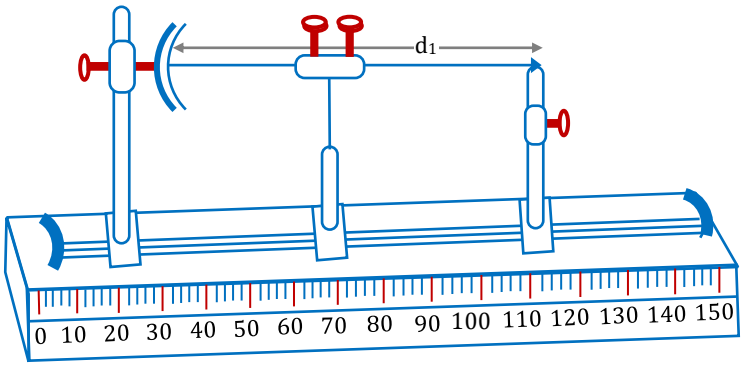
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Index Error = Observed distance – Actual distance and

Index Correction = Actual – Observed distance

Note:

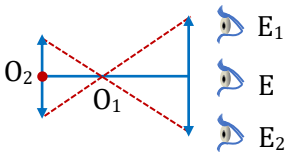
Index correction whether positive or negative, is always added algebraically to the observed distance to get the corrected distance.



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Parallax :

When two objects O_1 and O_2 are placed in such a way that both of them lie in the same line of sight as shown in figure, then the object nearer to the eye covers the object farther from it. Their images on the retina are superimposed and therefore, it is impossible to decide which is the nearer object. To identify this fact, the observer displaces his eye to a position E_1 or E_2 until he is able to see two distinct objects.

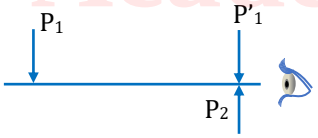


The more distant object O_2 apparently moves in the direction opposite to the displacement of the observer's eye with respect to the nearer object O_1. This relative shift in the position of two objects due to the shift in the position of the observer's eye is called parallax.

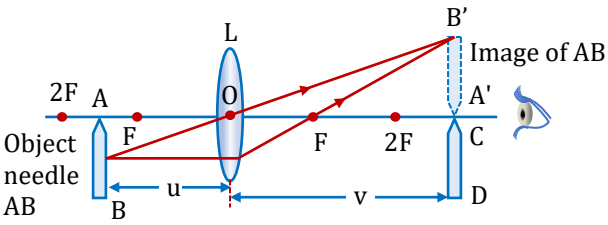
Parallax between the two objects disappear if they are at the same position.

The figure shows the tips of two pins P_1 and P_2 kept in the upright positions. The parallax between P_1 and P_2 is removed by shifting the position of observer's eye sideways. As the farther pin P_1 is displaced towards the pin P_2 the relative shift (parallax) between their positions decreases as the position of eye is displaced sideways. The relative shift vanishes when the pin P_1 occupies the position P'_1, that is, when the tips of the two are just coincident. At this position there is no parallax between the tips of the two pins.

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Experiment



Ray Diagram

Objective: To find the values of v for different values of u in case of a convex lens and to find its focal length.

Apparatus required: An optical bench with three uprights, one lens holder, two needles, given convex lens etc.

Formula required: Focal length of a lens from lens formula

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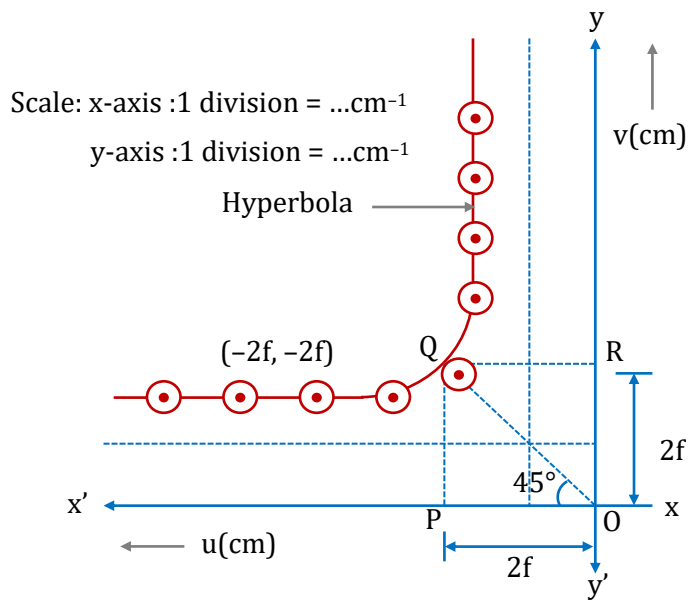
Procedure:

1. Roughly find the focal length of convex lens by focusing a sharp, clear and inverted image of a distant object on a white paper and measuring this distance between the lens and the white paper with a meter scale.
2. Level the optical bench. Mount the convex lens on the central upright of the optical bench. On the remaining two uprights mount the two needles. Arrange the tips of the needle at the same vertical height as the centre of the lens.
3. One needle *AB* is object needle and the other one *CD* is image needle and to make difference between them tip of one of the needles with a piece of chalk or putting a paper flag on it.
4. Shift the position of the object needle *AB* to a distance greater than 2*f* from the lens. Look from the other side of the lens along its principal axis near the end of the bench. If the setting is correct, an inverted, real image *A'B'* is seen. Now adjust the position of the second needle *CD* such that parallax between the image of the object needle and the image needle is removed.
5. Note the positions of the lense, the object needle and the image needle on the bench scale and thus find the observed values of *u* and *v*.
6. Repeat the above steps for 5 different positions of the object by placing it beyond 2 *F* and between *F* and 2*F*.

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Calculation from graph:

Method-I

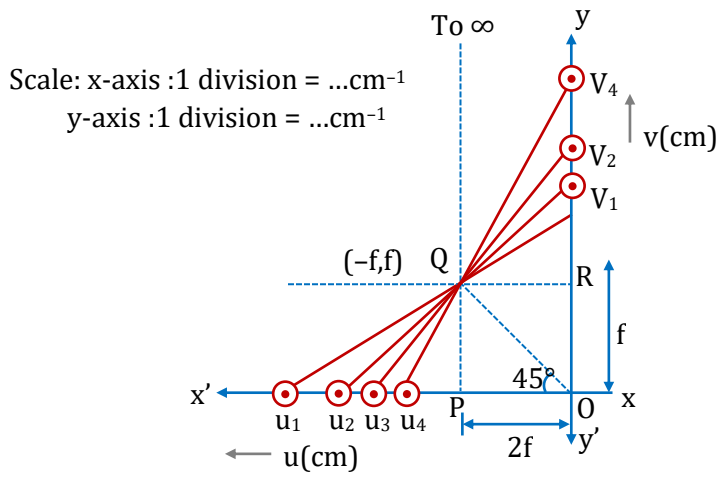


In this graph

Focal length $f = \frac{QP}{2}$ or $\frac{QP}{2}$

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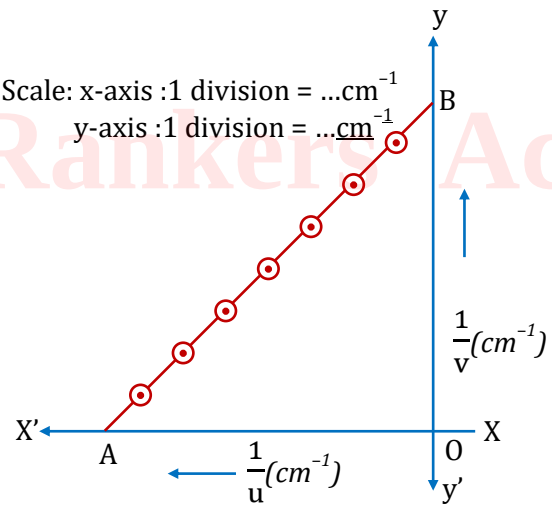
Method-II



In this graph,

Focal length $f = -QP = QR$

Method-III



In this graph,

Focal length $f = \frac{1}{OA} = \frac{1}{OB}$

Precautions:

1. All the uprights should be vertical.
2. The tip of the needles and centre of lens should be at same height.
3. Principle axis of the lens should be horizontal and parallel to the central line of the optical bench.
4. Parallax of the image and object needles should be removed tip to tip.
5. The object needle should be placed at such a distance that only real, inverted image of it will formed.

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Sources of Error:

1. The uprights may not be vertical.
2. Parallax removal may not be perfect.

Refractive index using travelling microscope:

Travelling Microscope:

It is a compound microscope fitted vertically on a vertical scale. It can be moved up and down. It consists of a vernier scale moving along the main scale.

The reading is taken by combining the main scale and vernier scale reading.

Experiment:

Objective: To determine the refractive index of a glass slab using a travelling microscope.

Apparatus: A marker, glass slab, travelling microscope, lycopodium powder, etc.

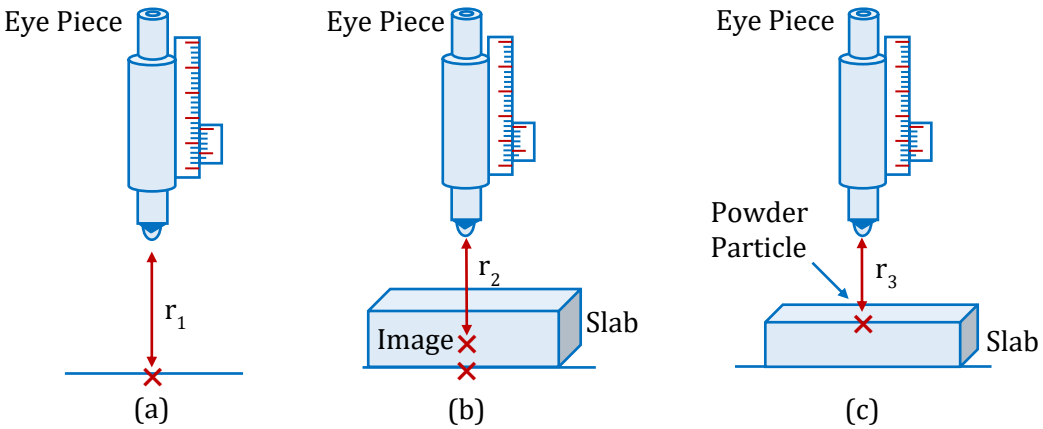
Concept: As, Refractive index,

$$\mu = \frac{\text{Real depth}}{\text{Apparent depth}}$$

To measure real and apparent depth a travelling microscope is used.

If the reading microscope when focussed on ink mark on white paper r_1, reading when the slab is kept over the ink mark is r_2 and the reading of image of lycopodium powder then real depth $= r_3 - r_1$ and apparent depth $= r_3 - r_2$. Therefore, refractive index of the material of slog is $\boxed{\mu = \frac{r_3 - r_1}{r_3 - r_2}}$.

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Procedure:

1. Note the number of divisions of vernier which coincide with number of main scale divisions. Find the value of each main division and hence least count of the microscope scale as (1 M.S.D.– V.S.D.).
2. Set the microscope in its stand such that it is capable of sliding vertically up and down as the screw attached to rack and pinion is turned.

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3. Mark a cross on a sheet of paper and place it below the objective of the microscope. Move the microscope very gently. Using the screw, focus the eye piece on cross mark and bring the cross in focus such that the cross of cross wires, coincides with the marked cross on the paper. Note the reading of the microscope as r_1.
4. Place the given glass slab on the cross mark. The cross mark appears to be raised.
5. Move the microscope gradually and gently upward to bring the cross mark in focus and on cross of cross wires. Record the reading as r_2.
6. Sprinkle some fine powder on the glass slab and move the microscope upward till the powder particles come into focus. Record the reading on the scale as r_3.
7. $r_3 - r_1$ = real depth and $r_3 - r_2$ = apparent depth.

Precautions:

1. Microscope should focus properly and reading should be taken carefully.
2. Extremely small amount of powder should be used.

Source of error:

1. Microscope may not be focused well.
2. It may have faulty calibration.

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